UTKARSH GARG

Tech Lead | Senior Full-Stack Engineer — Web Platforms, Distributed Data & Applied GenAI
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SUMMARY

Tech Lead with **6+ years** at D. E. Shaw building the web platform and developer infrastructure used firm-wide, currently owning four concurrent initiatives: a **GenAI coding assistant with 250+ daily active users**, the backend connector and compute engine of an in-house BI library used by **30+ teams**, a usage-logging platform serving **300+ applications** at **1.5M log entries/day**, and framework-level observability. Track record of proposing and driving flagship architecture to completion, delivering order-of-magnitude performance wins (**60x** via distributed compute, **138x** via Apache Arrow), leading firm-wide migrations end-to-end, and growing engineers to promotion. Deep in TypeScript, React, Node.js, Python, distributed systems, and LLM agent architecture.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES2020+), Python 3.11, SQL, C++, Java, HTML, CSS

Frontend: React 18, Redux/Reselect, React Query, React Router, Material UI, Storybook, Socket.IO, React Testing Library, Cypress

Backend & Distributed Data: Node.js, Express, FastAPI, REST API design, Dask, Apache Arrow, pandas, Apache Kafka, MongoDB, SQL, ClickHouse, DAG/compute-graph query engines, multi-tier caching

GenAI & Agents: LangChain, LangGraph, Claude Agent SDK, MCP servers, RAG, Milvus, FAISS, embeddings, Ragas, LLM-as-judge evaluation, prompt & agent orchestration

Infrastructure & Observability: Kubernetes, K9s, Docker, OpenTelemetry (tracing/metrics), Grafana, Core Web Vitals, OIDC/OAuth2 (on-behalf-of), Kerberos/SPNEGO, Linux

Tooling & Practices: Git, GitHub Actions (CI/CD), pnpm, JupyterLab, Cursor, Claude Code, VS Code, static security scanning, release management, PRD authoring

PROFESSIONAL EXPERIENCE

D. E. Shaw & Co. | Hyderabad, India Jul 2020 – Present

Tech Lead, Core Tech (Web Infrastructure) *Promoted: Sr. Member Technical 2022 • Tech Lead Dec 2024*

Own four concurrent platform initiatives on the Web Infrastructure team serving the firm-wide engineering org. Manage **2 direct reports** and mentor **5+ engineers** across full-stack web, distributed data, and GenAI.

GenAI Assistant & Developer Productivity Infrastructure

- Architected and shipped an agentic AI coding assistant — specialized in the firm's internal web framework and Python data-science APIs — serving **250+ daily active users** (~700 unique users, **1,500–2,000+ queries/week**), scaled from **~23 weekly users** at launch across internal docs, IDEs (Cursor, Claude Code, VS Code, Jupyter) and the in-house chat platform.
- Engineered an extensible multi-tool agent on **LangChain, LangGraph and the Claude Agent SDK** with integrated **MCP servers**, replacing a low-quality RAG v1 and delivering **30–40% higher accuracy** on evaluation benchmarks.
- Designed the three-tier retrieval stack: an MCP server exposing the assistant to IDEs, a RAG skill service, and versioned **Milvus** documentation collections plus **FAISS** stores over historical support threads, with query decomposition and re-ranking for higher answer precision.
- Built the evaluation, benchmarking and human-feedback framework from scratch (**Ragas** metrics, string match, LLM-as-judge, AST comparison, Cypress end-to-end), enabling regression detection and data-driven model/prompt decisions.
- Consolidated disparate services into a **unified Kubernetes hosting model** with usage, cost and token tracking plus full observability, materially improving reliability and adoption.
- Created the team's **first Claude plugin marketplace** and shipped a framework-aware **LLM code-review plugin** (full review, PR review, React review) with modular, owner-maintained rules codifying engineering best practices.

BI Library — Backend Connector Ecosystem & Compute Engine

- Own the backend connector ecosystem of the firm's in-house Tableau/Power BI equivalent, used by **30+ application teams** to build dashboards for traders, analysts and researchers.
- Proposed and drove to completion a ground-up **connector re-architecture** — gathered requirements across teams, then separated concerns so connectors own only data access while the engine owns orchestration (filter merging, query splitting, post-processing, deduplication of concurrent identical queries), enabling multi-dataset connectors with zero changes to the engine core.

- Architected a **DAG-based compute engine**: a shift from fixed-order stages to a topologically-sorted compute graph (filter / aggregate / evaluate / transform / sort) supporting on-the-fly histograms, nested-expression and rollup resolution, calculated fields, bucketing and intermediate-result caching — eliminating redundant computation.
- Re-implemented the data connector layer on **Dask with MapReduce patterns**, achieving a **60x performance improvement** on large datasets (25M rows: 10-minute timeouts → 10-second completions) and unlocking queries that previously OOM'd on pandas.
- Prototyped **Apache Arrow** as the columnar transport format, demonstrating **138x faster** UI property generation and **~68% payload reduction** for flat-table workloads.
- Designed and shipped **cross-filtering** end-to-end across 12–15 visualization types and **multi-dataset** support for cross-source aggregations; defined a three-tier caching strategy (client LRU with in-flight dedup, server, OLAP).
- Built a **JupyterLab** integration auto-generating interactive UIs from pandas DataFrames, closing the gap between the JS library and the firm's Python-first research workflows.

Usage Logging & Observability Infrastructure

- Led the solo, end-to-end migration of the usage-logging backend from **MongoDB to SQL**; onboarded **80% of the firm's 300+ applications** within a year (**1.5M log entries/day**), met a firm-wide **Kerberos authentication** deadline, decommissioned the legacy stack, and initiated a **ClickHouse** evaluation for long-term performance.
- Identified a critical observability gap in the internal web framework powering **300+ business-facing applications** and drove first-class observability as a built-in capability, eliminating per-team telemetry work.
- Architected an out-of-the-box stack — distributed tracing, client-side Core Web Vitals, server metrics and error aggregation — configurable through a **single observability key** and shipped across two deployment models; coordinated with Python Infra to stitch **OpenTelemetry** traces across polyglot JS + Python services.
- Roadmap owner (2025) for observability, debuggability and usage analytics; rolled out Grafana dashboards and alerting on by default, plus an out-of-the-box usage-analytics dashboard giving 300+ applications instant visibility with zero custom setup.

Firm-Wide Shared UI Library (Discussion & Ticketing Toolkit)

- Sole technical owner of a full-stack JavaScript library providing embeddable discussion, request and ticketing UI across the platform; drove it from **v13 to v26** over six years, owning every release, migration guide and consumer escalation.
- Led a full **design-system migration of 80+ React components** to the firm's MUI-based library and migrated the real-time backbone from **ActiveMQ/STOMP to Apache Kafka** with a permission-gated Socket.IO subscription manager; implemented **OIDC on-behalf-of** authentication replacing username-passing with per-user bearer tokens.
- Executed toolchain modernizations with consumer migration guidance — React Router v5→v6, React 18, Node 16→20, pnpm — and maintained CI running builds, docs and Cypress integration suites against ephemeral app instances with static security scanning.

Leadership, Hiring & Developer Advocacy

- Mentored **5+ junior engineers** (**2–3 promoted**); interviewed **100+ candidates** since 2022, hiring for Tech Lead roles and representing the firm at **IIT/NIT** campus drives.
- Member of the org's **Productivity Committee**; delivered BU-wide lightning talks on Claude Code workflows, plugins and automation, and built a developer daily-summary tool aggregating ticketing, GitHub, Confluence and Outlook activity.
- Led the **2025 Web Developer Productivity Survey**, improving measurement methodology and surfacing **87% AI-tooling adoption, 95% reporting productivity gains, NPS 57.6** — used to shape platform roadmap.

Visa Inc. | Bangalore, India May 2019 – Jul 2019

Software Engineering Intern

- Built a **React/Redux + GoLang** status tracker replacing the team's email-based SDLC tracking, including GoLang endpoints for tracker configuration and data workflows.

EDUCATION

Motilal Nehru National Institute of Technology (MNNIT), Allahabad 2016 – 2020

B.Tech, Computer Science & Engineering • **CGPA: 9.16 / 10**

ACHIEVEMENTS

- Promoted to **Senior Member Technical in 1.5 years** as a top performer, then to **Tech Lead** (Dec 2024).
- **Merit Scholarship** recipient at MNNIT Allahabad — top 18 of 176 students for academic performance.
- **Special Mention**, D. E. Shaw Data Science League.